Roman Knyazhitskiy

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Summary

Research Engineer at Mirai Labs, enjoying efficient ML, learning theory and robotics. My side interests include PL theory, philosophy of science and AI Safety.

Education

MPhil in MLMI (Machine Learning), University of Cambridge 10/2025 – 09/2026

BSc Computer Science and Engineering + Honours, TU Delft 09/2022 – 07/2025

Work Experience

Research Engineer, Mirai Labs 01/2026 – now

- Worked on post-training quantization, speculative decoding, continuous batching and automatic batch-size estimation.
- Led the efforts to setup proper evaluation pipelines for our quantized models.

Research Associate, TU Delft 03/2023 – 08/2025

- Developed a transformer-based Prior-Data Fitted Network (PFN) for hyperparameter optimization, achieving 15–20% improvement over the current SOTA.
- Presented a deep dive on the modern view on first-order oracle optimizers (Bernstein et al.), showing its connection to proximal gradient descent.

Student Team ML Lead, Delft Mercurians 05/2023 – 09/2025

- Led machine learning for an autonomous robot soccer team (RoboCup Small Size League).
- Deployed a real-time Model Predictive Control system on hardware (15 ms latency), interfacing Python with Rust via PyO3.

Projects

- **Continuous-time Differentiable Simulator (2023):** Implementation of a small BRAX-like physics engine in JAX, with simple solvers and better multi-body support.
- **Stack Associated Beam Tracing (2022):** Likely saymptotically fastest octree tracing/rendering algorithm at the time.
- **Robotics (2015-2026):** I've been writing code for various robots and embedded systems for 11 years now, mostly in C/C++. Recently moved on to more applied ML side.

Honours and Awards

- **1st Place**, Bunq Hackathon 6 (2025); €30,000 prize
- **2nd Place & Special Prize**, Epoch AI Hackathon (2024)
- **Best Software Solution**, RoboCup World Championships, Sydney (2019)

Other

- Contributed to jaxtyping, Equinox and beartype stack, attempting to improve runtime typechecking for users of IPython notebooks.
- Was a member of Meridian, the AI Safety Hub in Cambridge; indirectly contributed to some work on preventing Evaluation Awareness in LLMs.

Publications

[1] J. Luijmes, A. Gielisse, R. Knyazhitskiy, and J. van Gemert. ARC: Anchored representation clouds for high-resolution INR classification. In *ICLR 2025 Workshop on Weight Space Learning*, 2025. Accepted.

[2] Roman Knyazhitskiy and Andrea Giuseppe Di Francesco. Early-exit graph neural networks for link prediction. Accepted at LoG@Pisa, 2026.

[3] R. Knyazhitskiy and P. R. Van der Vaart. A simple scaling model for bootstrapped DQN, 2025. Under review.